THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The curved surface area of a hemisphere of radius r is: 1

(a) πr^2 **(b)** $4\pi r^2$ **(c)** $3\pi r^2$ **(d)** $2\pi r^2$

2. The volume of a cube is 512 cm^3. Its edge is: 1

(a) 9 cm **(b)** 8 cm **(c)** 7 cm **(d)** 10 cm

3. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 7 cm and height 6 cm is: 1

(a) 924 cm^3 **(b)** 1017 cm^3 **(c)** 1110 cm^3 **(d)** 831 cm^3

4. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 14 cm is: 1

(a) 2958 cm^2 **(b)** 2711 cm^2 **(c)** 2464 cm^2 **(d)** 2217 cm^2

5. The volume of a cuboid of length 5 cm, breadth 8 cm and height 9 cm is: 1

(a) 360 cm^3 **(b)** 323 cm^3 **(c)** 434 cm^3 **(d)** 397 cm^3

6. The volume of a cube of edge 2 cm is: 1

(a) 11 cm^3 **(b)** 8 cm^3 **(c)** 9 cm^3 **(d)** 7 cm^3

7. The total surface area of a cube of edge 6 cm is: 1

(a) 261 cm^2 **(b)** 216 cm^2 **(c)** 194 cm^2 **(d)** 238 cm^2

8. The lateral surface area of a cuboid 3 cm $\times$ 4 cm $\times$ 5 cm (height 5 cm) is: 1

(a) 62 cm^2 **(b)** 70 cm^2 **(c)** 86 cm^2 **(d)** 78 cm^2

9. In a cone, the relation between radius r, height h and slant height l is: 1

(a) $r^2 = l^2 + h^2$ **(b)** $l^2 = r^2 + h^2$ **(c)** $h^2 = r^2 + l^2$ **(d)** $l = r + h$

10. The volume of a cube is 343 cm^3. Its edge is: 1

(a) 7 cm **(b)** 6 cm **(c)** 8 cm **(d)** 9 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A water tank is 8 m long, 5 m wide and 9 m deep. How many litres of water does it hold when full? (1 m³ = 1000 litres) 2

12. Find the volume, total surface area and lateral surface area of a cuboid of length 8 cm, breadth 9 cm and height 6 cm. 2

13. Find the volume and total surface area of a cube of edge 8 cm. Also find the edge of a cube whose volume is 729 cm³. 2

14. A water tank is 9 m long, 6 m wide and 7 m deep. How many litres of water does it hold when full? (1 m³ = 1000 litres) 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the volume, total surface area and lateral surface area of a cuboid of length 7 cm, breadth 14 cm and height 9 cm. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 3.5 cm and height 4 cm. 3

17. Find the volume and total surface area of a cube of edge 20 cm. Also find the edge of a cube whose volume is 9261 cm³. 3

OR

Find the volume, total surface area and lateral surface area of a cuboid of length 8 cm, breadth 12 cm and height 11 cm. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 14 cm and height 10 cm. 3

OR

Find the volume and total surface area of a cube of edge 13 cm. Also find the edge of a cube whose volume is 2744 cm³. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 21 cm and height 11 cm. 5

OR

How many cubes of edge 3 cm can be cut from a cuboid measuring 9 cm × 12 cm × 6 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

20. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 10.5 cm and height 18 cm. 5

OR

How many cubes of edge 4 cm can be cut from a cuboid measuring 16 cm × 24 cm × 8 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

END OF QUESTION PAPER — **Total: 40 Marks**